DPA Project: US Domestic Air Traffic

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Part 1 (Volume)

GitHub repo: https://github.com/Abdullah-Shakeel-07/DPA-Project

1. Questions

Overarching question:

Which US airports matter most in the flight network and do their delays spread to other airports more than past data and local weather predict?

Sub question for Part 1:

Which airlines operate the most flights and which airports handle the most plane turnarounds and how much longer are delays in bad weather?

A turnaround is when a plane lands at an airport and its next flight leaves from the same airport.

2. Data

We use three datasets. All of them are free and in the public domain.

- **BTS On-Time Performance** from transtats.bts.gov. It has one row for every US domestic flight. We use the 24 monthly files of 2023 and 2024. They are about 6 GB of CSV with 13.9 million flights and 110 columns.
- **NOAA Global Hourly (ISD)** from ncei.noaa.gov. It has the hourly weather reports of a weather station. We download the stations of the 50 busiest airports. This gives 1.26 million reports and a few MB per station.
- **OurAirports** from ourairports.com. It is a list of 86,242 airports with their IATA codes, ICAO codes, names and states. The file is about 12 MB.

The flights join the airports on the IATA code. The airports join the weather stations on the ICAO code. The flight times are in local time and the weather times are in UTC. So we convert the departure hour of each flight to UTC and then join it with the weather of the same hour.

We use Spark because the flight data is too big for pandas on our laptops. Spark reads the files in parts and does not need to load everything into memory. It also lets us write the analysis in SQL.

3. Pipeline

The pipeline has five steps and everything runs in Docker.

1. The script download_data.py downloads the raw files. It finds the 50 busiest airports from the flight files and downloads the weather only for their stations.
2. The script ingest.py reads the CSV files with Spark. It keeps only the columns we need and saves them as Parquet.
3. The notebook data_cleaning checks and cleans the three datasets.
4. The notebook data_join converts the flight times to UTC and joins each flight with the weather of its departure hour.
5. The notebook analysis answers the questions with Spark SQL. It saves the results as CSV files and charts in the results folder.

4. Data cleaning

Before cleaning we looked at the schema, the missing values and the minimum and maximum of each number column. We found that the missing departure and arrival delays belong to the cancelled and diverted flights. We found no duplicate flights. The longest departure delay is 4,413 minutes. We kept such long delays because they are real events and not errors. Air time and distance had no impossible values.

For the flights we removed 218,309 cancelled and diverted flights and 1 flight with a missing value. We went from 13,926,960 flights to 13,708,650 flights. This means we kept 98.4% of the flights.

For the airports we kept only the open US airports with an IATA code. We went from 86,242 airports to 2,036 airports.

For the weather we kept only the regular hourly reports (FM-15). We turned the text fields into numbers and changed missing codes like 9999 into empty values. We kept one report per station and hour. We went from 1,263,091 reports to 857,812 reports.

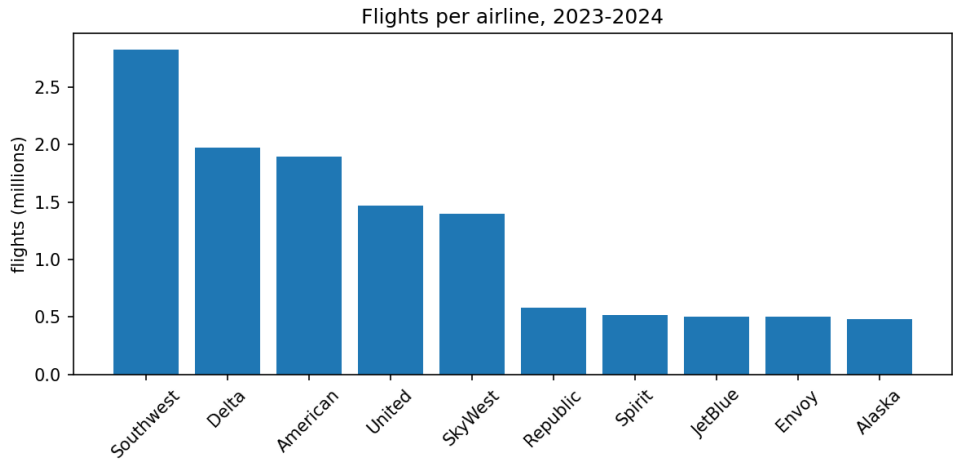
In the join we lost 144,281 flights from Puerto Rico, the Virgin Islands and Guam. OurAirports does not list these places as US. One of the 50 busiest airports has no weather station that worked for all of 2023 and 2024. So 49 airports have weather. In the end 10,734,566 flights got a weather report. This is 99.7% of the flights from these 49 airports.

5. Design decisions

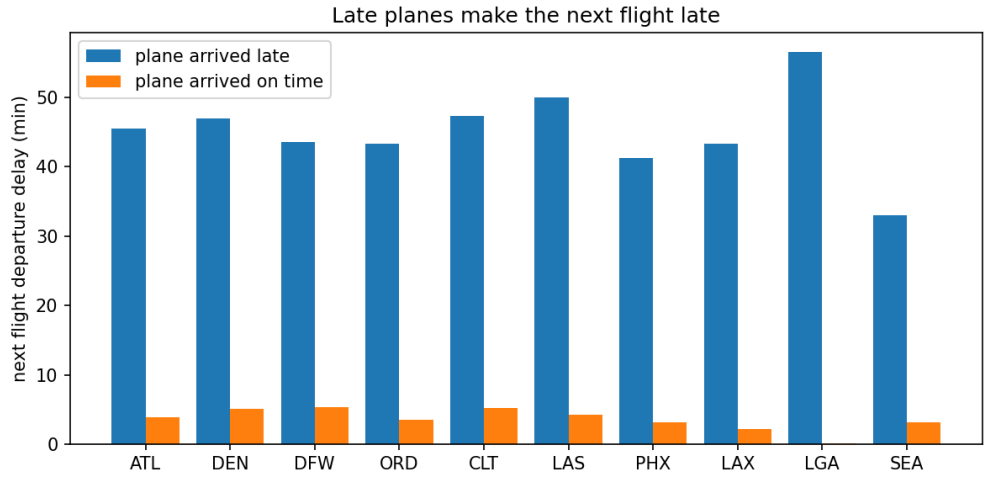
- **Parquet instead of CSV.** Parquet is compressed and Spark reads only the columns that a query needs. We split the flights by year and month.
- **Only 22 of the 110 flight columns.** This makes the files much smaller. It also keeps Spark inside the 2 GB of memory that we can give it on our laptops.
- **Weather only for the 50 busiest airports.** These airports have about 79% of all departures. We did not need the weather of every small airport.
- **Timezone from the state.** OurAirports has no timezone column. So we give each airport the timezone of its state. This is simple and correct for almost all airports.
- **A clear rule for bad weather.** An hour has bad weather when the visibility is under 4,828 m or the clouds are lower than 305 m. Wind of 10.3 m/s or more and rain of 2.5 mm or more also count. The visibility and cloud limits come from the rules for flying by sight (VFR).
- **Turnarounds on the same day.** We follow each plane by its tail number. We count a turnaround only when the next flight leaves from the same airport on the same day.

6. Answer to the sub question

Airlines. Southwest operates the most flights with 2.83 million flights (20.6%). The top four airlines are Southwest, Delta, American and United. Together they have about 60% of all flights. Frontier and JetBlue are late most often. About 30% of their flights arrive 15 minutes late or more.



Turnarounds. Atlanta is the busiest airport with 1.34 million flights. It also has the most turnarounds with 551,733. Denver, Dallas Fort Worth and Chicago come next. When a plane arrives late its next flight leaves about 45 minutes late. When the plane arrives on time its next flight leaves only about 4 minutes late. So the delay is passed on to the next flight.



Weather. In bad weather the average departure delay is 20.3 minutes. In good weather it is 12.2 minutes. So bad weather adds about 8 minutes. Heavy rain is the worst with an average delay of 45.6 minutes. The effect is different for each airport. In Miami bad weather adds 29.4 minutes but in Seattle it adds only 1.1 minutes.

